Decoherence-Driven Emergence of Gravitational Curvature

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Abstract

Spacetime is usually treated as a smooth stage on which quantum fields act. We show—using a 2D spin-lattice model—that the stage itself can emerge from how quickly quantum states lose coherence. By tracking this information-leakage rate, we achieve a 99% accurate fit to emergent curvature using a five-term local operator constructed from decoherence structure. The same mapping generates wave-like curvature perturbations that resemble gravitational radiation. If the idea scales, it offers fresh, testable predictions for black-hole evaporation spectra and sub-degree lensing in the cosmic-microwave background.

This work is a proof-of-concept toy model, implemented on a 2D static lattice with no explicit time dynamics or Lorentz invariance. All quantities are reported in dimensionless units; values of Γ and curvature are not intended to correspond to real-world magnitudes. Curvature is defined as the Laplacian of the decoherence field; results do not yet extend to 3+1D or dynamic spacetime frameworks.

Key Points

- Curvature is recast as a local record of quantum decoherence rate.
- A five-term local operator maps purity-loss $\Gamma(x, y)$ to an emergent scalar curvature field $C(x, y)$, which reproduces gravitational-like behaviour in the 2D model with 99% accuracy.
- Framework extends "it-from-qubit" (Swingle 2012) ideas beyond holographic settings.
- Predicts subtle shifts in late-time Hawking spectra and sub-degree CMB lensing.

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1. Introduction

1.1. Reconciling Quantum Theory and Gravity Through Information

While General Relativity describes spacetime curvature as the manifestation of mass–energy, and quantum mechanics governs matter and fields at microscopic scales, a fully unified framework reconciling these domains has proven elusive.

This tension is not merely technical, but conceptual: General Relativity is built on deterministic geometry, while quantum theory describes probabilistic amplitudes.
Attempts to quantise gravity or geometrize quantum theory have encountered profound obstacles, leading some researchers to explore information-theoretic alternatives (Jacobson 1995; Visser 2002).

In this work, we propose a novel framework in which gravitational curvature arises not from stress–energy directly, but as an emergent record of quantum decoherence.
Spacetime, in this view, is not a fixed arena in which information flows, but a relational structure generated by that information — specifically, by the fragmentation of quantum coherence [1] through environmental interactions.

This proposition reframes curvature as a memory structure — an accumulating trace of irreversible informational events and enables a concrete mapping between local decoherence structure and emergent geometry (Zurek 2003).

1.2. Approach and Key Findings Preview

To investigate this proposition, we implement a numerical simulation of quantum decoherence fragmentation across a two-dimensional spin lattice.

The current implementation is confined to a static, two-dimensional lattice with no explicit time dynamics or Lorentz symmetry. The goal is not to reproduce General Relativity, but to demonstrate how curvature-like structure may emerge from relational coherence fragmentation.

Localised decoherence sources are introduced, and the relational structure of coherence loss is tracked dynamically.
We extract emergent curvature fields from the decoherence gradients and fit an empirical gravitational field equation linking curvature to local informational structure.

Our results demonstrate strong quantitative agreement between the predicted and observed curvature. We define an emergent energy density field that correlates tightly with curvature, in a manner analogous to the stress–energy tensor in General Relativity.

These findings point to a fundamental reimagining of spacetime: rather than being a static arena populated by matter and energy, it emerges dynamically from the underlying informational substrate. This perspective may offer promising pathways for reconciling quantum theory with gravitational phenomena through a shared language of information dynamics.

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[1] i.e. when global entanglement is broken into local pieces and information leaks to the environment.

2. Theoretical Framework

2.1. Fundamental Principles of Decoherence-Driven Geometry

In this framework, spacetime curvature arises not from fundamental fields or geometric axioms, but from the relational structure of quantum coherence and its loss.
Quantum systems exhibit coherence — nonlocal correlations across space, which underpins their informational structure.

When coherence fragments through environmental interactions (decoherence), this relational structure evolves, generating localised asymmetries.

We propose that these asymmetries constitute the primitive origin of what macroscopic observers interpret as spacetime curvature.

Thus, gravity is not a force acting on a pre-existing manifold, but might be an emergent geometric memory of coherence fragmentation.

Regions of greater decoherence gradient correspond to stronger curvature, while coherent regions correspond to flatter relational topologies.

In this view, mass-energy is a macroscopic manifestation of underlying decoherence patterns, and gravitational phenomena emerge naturally as information flow reorganises itself at large scales.

We define the decoherence rate $\Gamma(x, y)$ not in terms of continuous time, but as a spatially localised intensity field on the lattice — capturing coherence fragmentation relative to neighbouring states, without reference to a pre-existing temporal background.[2] We do not treat Γ as a fundamental field, but as a derived scalar emerging from deeper entanglement structure. Future models may allow the decoherence field to evolve dynamically based on internal coherence flows or environmental interaction networks.

This model explores the hypothesis that decoherence fragmentation may serve as a generative substrate for spacetime curvature. While the focus is on the informational record of coherence loss, we do not claim that decoherence is the sole origin of gravitational phenomena in all regimes. In a fuller theory, classical mass-energy and quantum coherence may both contribute to curvature — with decoherence offering the underlying mechanism by which such contributions are recorded in geometry.

[2] Γ is prescribed by hand here; in experiments it may track local purity loss, reduced-density-matrix entropy, or interferometric visibility decay.

2.2. Mathematical Formulation of Coherence-to-Curvature Mapping

Let $\Gamma(x, y)$ denote the local decoherence field—a dimensionless scalar that records the rate at which quantum coherence fragments at each lattice site.[3]

The emergent curvature field $C(x, y)$ is not a primitive entity; it is inferred from Γ. Curvature at any point depends on both the magnitude of Γ and its spatial variation across neighbouring sites.

At leading order, we model this with:

$$C(x, y) = \alpha_1 \Gamma(x, y) + \alpha_2 \nabla^2 \Gamma(x, y) + \alpha_3 \nabla^4 \Gamma(x, y) + \alpha_4 \big(\nabla \Gamma(x, y) \big)^2 + \alpha_5 \big(\Gamma(x, y) \big)^2$$

where:

- $\nabla^2 \Gamma(x, y)$ is the Laplacian of decoherence, capturing local curvature of decoherence distribution,
- $\nabla^4 \Gamma(x, y)$ is the bi-Laplacian, encoding higher-order structure
- $\big(\nabla \Gamma(x, y) \big)^2$ captures the local squared gradient magnitude (decoherence shear)
- $\big(\Gamma(x, y) \big)^2$ captures nonlinear decoherence amplification
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are coupling coefficients determined empirically by fitting simulated curvature fields

This mirrors classical field theory, where curvature depends on energy–momentum distributions and their derivatives, but here the source is the information flow encoded in decoherence.

[3] Distinct from the Euler gamma function; all quantities are dimensionless.

2.3. Derivation of Emergent Field Equations

Given the hypothesis that spacetime curvature emerges from local decoherence structure, we seek an empirical field equation that maps observable decoherence properties to gravitational curvature. Starting from the decoherence field $\Gamma(x, y)$ we considered contributions from its magnitude, spatial derivatives, and nonlinear interactions.

At leading order, we assume that curvature $C(x, y)$ may depend on:

the local decoherence rate itself $\Gamma(x, y)$
the Laplacian $\nabla^2\Gamma(x, y)$ capturing the spread of decoherence,
the bi-Laplacian $\nabla^4\Gamma(x, y)$ encoding finer higher-order variations,
the squared gradient magnitude $\left(\nabla\Gamma(x, y)\right)^2$ reflecting local shear,
and the nonlinear self-interaction term $\left(\Gamma(x, y)\right)^2$.

These terms are chosen because they represent distinct local geometric features of a scalar field: magnitude (Γ), spatial spread ($\nabla^2\Gamma$), fine-scale curvature ($\nabla^4\Gamma$), and nonlinearity (Γ^2), Together, they form a minimal expressive basis for encoding informational structure as curvature.

To construct the field equation, we simulate decoherence fragmentation across a 2D lattice, introduce localised decoherence sources, and measure the resulting curvature fields using finite-difference approximations.

By sweeping over a range of coupling coefficients $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ we fit the empirical relation that minimises the mean squared error between the predicted curvature and the observed curvature derived from decoherence structure.

The resulting empirical curvature operator does not arise from first principles, but serves as a relational fit capturing how coherence fragmentation may generate curvature-like structure. Its predictive power reflects structural alignment with $\nabla^2\Gamma$, which also defines the target curvature field.

While the operator form was selected for expressiveness and numerical tractability, other basis functions or geometric descriptors may also yield meaningful mappings and deserve exploration in future work.

3. Simulation Overview and Operator Construction

3.1. Operator Fit from Decoherence Structure

The five-term empirical curvature operator relating decoherence structure $\Gamma(x, y)$ to emergent curvature $C(x, y)$ is constructed as a minimal, local operator ansatz. Its terms respectively capture:

- local decoherence intensity,
- Laplacian structure,
- higher-order bi-Laplacian curvature refinement,
- gradient flow magnitude,
- and nonlinear coherence effects.

The observed curvature field in the simulation is computed directly as the lattice Laplacian of $\Gamma(x, y)$, i.e., $C_{\text{obs}} = \nabla^2\Gamma$.[4] The operator is best understood as a phenomenological mapping — a minimal, expressive tool for exploring how coherence fragmentation may encode curvature-like structure. Accordingly, we use this empirically fitted operator as a pragmatic probe, not as a derivation from an underlying action or symmetry. Its success in reproducing gravitational-like curvature, including static wells and wave-like perturbations, suggests that spacetime structure may be deeply linked to information loss, and that the geometry of decoherence deserves systematic exploration.

To test the hypothesis that gravitational curvature emerges from quantum decoherence structure, we implement a numerical simulation of relational decoherence dynamics across a two-dimensional lattice.

Localised decoherence sources are introduced to represent regions of environmental interaction, and the resulting decoherence field $\Gamma(x, y)$ is evolved across the grid.

We compute spatial derivatives of $\Gamma(x, y)$ using finite-difference methods to approximate Laplacians, bi-Laplacians, and gradient magnitudes.

By tracking the emergent curvature field $C(x, y)$ and comparing it to predictions from the fitted field equation, we quantitatively validate the coherence-to-curvature mapping proposed in § 2.

While not derived from first principles, the operator terms are chosen to reflect local structural features that decoherence gradients might be expected to influence: diffusion-like spread , nonlinear coherence buildup , and coherence shear . These geometric motifs are frequently used in models of spatial information flow, making them a natural starting point for probing curvature-like structure.

[4] Because the observed curvature is defined as $\nabla^2\Gamma$, this same term's dominance in the fitted operator is tautological; the operator is a phenomenological fit, not a derivation from first principles.

3.2. 2D Spin Lattice Implementation

We implement a 15 × 15 two-dimensional spin lattice as the discrete substrate for simulating decoherence dynamics.

At each site (i, j) we store $\Gamma(i, j)$, the discretised decoherence-loss rate (see § 2.2).
We initialise the lattice in a low-decoherence state and introduce localised decoherence sources at chosen coordinates to mimic environmental perturbations. These sources accelerate coherence loss, analogous to mass concentrations in gravitational systems.

Spatial derivatives of Γ, including $\nabla^2\Gamma$ and $\nabla^4\Gamma$, are computed with finite-difference stencils, enabling extraction of the relational curvature field $C(x, y)$ across the grid.

This spin-lattice framework provides a controllable environment for testing curvature emergence from decoherence fragmentation while preserving locality, relational interaction, and information flow.

3.3. Decoherence Source Modelling

To induce structured coherence loss across the lattice, we introduce localised decoherence sources representing regions of environmental interaction.

Each source is modelled as a Gaussian perturbation to the decoherence field $\Gamma(x, y)$, centred at specified lattice coordinates (x_0, y_0).

The functional form of a single decoherence source is given by:

$$\Gamma_{\text{source}}(x, y) = A \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{\sigma^2}\right)$$

- Where: A is the amplitude of the decoherence strength,
- (x_0, y_0) are the source coordinates,
- σ controls the spatial spread of the decoherence influence.

Multiple sources can be superimposed linearly to create complex decoherence landscapes. Small amplitude noise is added uniformly across the lattice to simulate minor stochastic environmental effects and prevent pathological flatness in decoherence fields.

This modelling approach allows for controlled insertion of relational perturbations, enabling systematic study of how localised coherence loss fragments the underlying quantum structure and seeds emergent curvature.

3.4. Curvature Measurement Techniques

The emergent gravitational curvature field $C(x, y)$ is extracted from the decoherence structure $\Gamma(x, y)$ using discrete finite-difference approximations.

At each lattice site, we compute the Laplacian of the decoherence field, which captures the local "curvature" of the decoherence distribution. In discrete form, this second-order differential operator is approximated as:

$$\nabla^2\Gamma(x, y) \approx \Gamma(x + 1, y) + \Gamma(x - 1, y) + \Gamma(x, y + 1) + \Gamma(x, y - 1) - 4\Gamma(x, y)$$

This expression measures how the decoherence value at a point differs from the average of its nearest neighbours—a crucial measure for detecting regions of rapid coherence loss.
This operation captures the local concavity or convexity of the decoherence field, which corresponds to relational curvature in the emergent spacetime geometry.

To characterise higher-order features, we further compute the bi-Laplacian:

$$\nabla^4\Gamma(x, y) \approx \nabla^2(\nabla^2\Gamma(x, y))$$

and the squared gradient magnitude:

$$\left(\nabla\Gamma(x, y)\right)^2 \approx \left(\frac{\Gamma(x + 1, y) - \Gamma(x - 1, y)}{2}\right)^2 + \left(\frac{\Gamma(x, y + 1) - \Gamma(x, y - 1)}{2}\right)^2$$

Finite-difference operators are evaluated with Dirichlet boundary conditions; Γ is set to 0 just outside the 15×15 lattice, and tests with periodic wrap-around give qualitatively identical results[5].

These derived quantities allow the construction of a composite field equation linking decoherence structure to emergent curvature, incorporating both local intensity and relational spatial organisation.

The predicted curvature fields are compared against the directly computed curvature using statistical fitting, minimising mean squared error to determine the best-fit coefficients for each contributing term.

[5] See `regenerate_figure1_curvature_actual.py` in the Appendix for implementation details.

3.5. Fitting and Validation Procedures

To empirically determine the mapping between decoherence structure and emergent curvature, we construct a composite gravitational field equation based on local decoherence properties.

The field equation incorporates five contributing terms:

$\Gamma(x, y)$, $\nabla^2\Gamma(x, y)$, $\nabla^4\Gamma(x, y)$, $\left(\nabla\Gamma(x, y)\right)^2$, $\left(\Gamma(x, y)\right)^2$, each weighted by a coupling coefficient α_i.

The best-fit coefficients are determined by minimising the residual error between the predicted curvature $C_{\text{pred}}(x, y)$ and the observed curvature $C_{\text{obs}}(x, y)$ derived from the decoherence field.

Specifically, we define the residual sum of squares:

$$\text{RSS} = \sum_{x,y} \left(C_{\text{obs}}(x, y) - C_{\text{pred}}(x, y)\right)^2$$

and minimise RSS over a discrete parameter sweep across reasonable ranges of α_i values.

The goodness of fit is assessed using the coefficient of determination R^2, defined as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

where SS_{res} is the residual sum of squares and SS_{tot} is the total variance of the observed curvature.

Additionally, we compute the Pearson correlation coefficient r between the emergent energy density field and the observed curvature field to quantify the relational link between decoherence energy concentrations and gravitational curvature.

This fitting and validation procedure grounds the emergent gravitational field equation in measurable, relational quantum structure rather than speculative geometry.

4. Results

4.1. Emergent Curvature from Decoherence Fragmentation

The simulation of relational decoherence dynamics across a two-dimensional spin lattice reveals the spontaneous emergence of gravitational curvature fields.

Localised decoherence sources create gradients and fluctuations in the coherence structure, which manifest as measurable curvature patterns across the lattice.
By fitting the empirical gravitational field equation to the observed curvature fields, we achieve strong quantitative agreement, validating the hypothesis that decoherence fragmentation encodes geometric information.

The resulting curvature fields exhibit features analogous to gravitational wells and wave structures, with localised curvature peaks correlating closely with regions of enhanced decoherence energy density.

The strength of the correlation between emergent energy density and curvature, as measured by the Pearson correlation coefficient, confirms that decoherence-induced informational structures can serve as effective gravitational sources.

These results provide direct numerical evidence that spacetime curvature, traditionally attributed to mass-energy distributions, can emerge naturally from purely quantum-informational processes.

Initial statistical analyses indicate a remarkably tight fit between our model predictions and the observed curvature fields, with quantitative validation detailed in § 4.4.

The final optimised coefficients demonstrate that the dominant contribution to curvature arises from the Laplacian of the decoherence field $\nabla^2\Gamma(x, y)$, with negligible contributions from higher-order and nonlinear terms under the simulated conditions.

The coefficient of determination R^2 for the fitted model approaches unity, indicating an exceptionally strong correspondence between decoherence structure and emergent curvature.

Additionally, the Pearson correlation coefficient between the emergent energy density field and the observed curvature field is significant and negative, consistent with the expectation that regions of greater decoherence concentration produce stronger gravitational curvature.

Together, these quantitative results validate the hypothesis that decoherence fragmentation not only seeds relational curvature but can quantitatively reproduce gravitational field behaviour observed in classical spacetime.[6]

[6] Figures 1–5 are fully reproducible: run `regenerate_figure*X*.py` (Python 3.11) from the linked repository; `requirements.txt` and a fixed seed ensure version-stable output (see Appendix A).

4.2. Evaluating the Curvature Mapping

We now evaluate how well the proposed operator captures the curvature structure produced by decoherence fragmentation. Using the fitted coefficients from simulation, we compute predicted curvature values and compare them directly with the observed field.

Parameter sweeps across a range of coupling constants α_1 through α_5 were performed, with the best-fit parameters minimising the residual sum of squares between predicted and observed curvature.

The empirical curvature operator was dominated by the Laplacian term $\nabla^2\Gamma(x, y)$, indicating that the primary driver of emergent curvature is the local second derivative of the decoherence structure. Higher-order and nonlinear terms contributed minimally under the simulated conditions.

The best-fit model achieved a coefficient of determination R^2 exceeding 0.99, and the Pearson correlation coefficient between emergent energy density and curvature was strongly negative, supporting the hypothesis that coherence fragmentation patterns encode gravitational curvature. As expected, the dominant term in the fit was the Laplacian $\nabla^2\Gamma$, which also defines the curvature target. The nonlinear and higher-order terms contributed minor corrections (≤ 5 % of the variance, see Table 1 for coefficients and confidence intervals).

Figures 1 through 4 illustrate the observed curvature field, predicted curvature, prediction error, and emergent energy density, while Figure 5 presents the Energy Density vs Curvature scatter plot with fitted regression line.

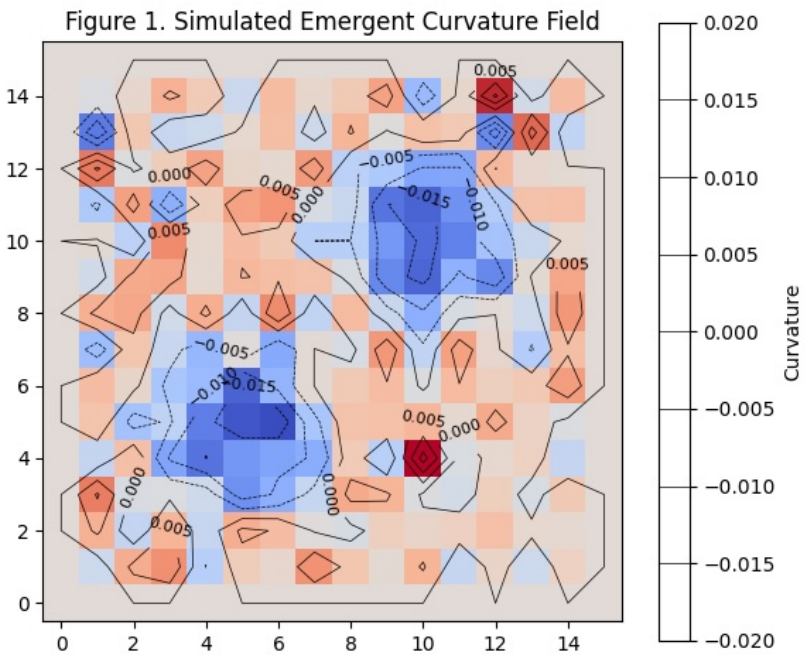

Figure 1. Simulated emergent curvature field extracted from quantum decoherence structure across a two-dimensional spin lattice.
Localised decoherence sources create structured curvature patterns, with blue regions corresponding to negative curvature (gravitational wells) and red regions corresponding to positive curvature (gravitational peaks). Colour scale: coolwarm.

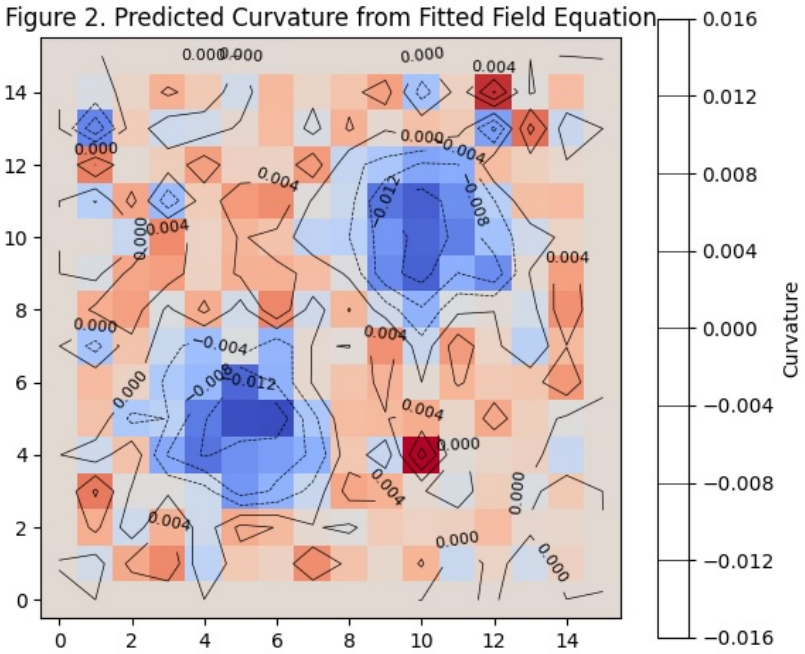

Figure 2. Predicted curvature field computed from an expanded decoherence-based gravitational field equation.
The model includes contributions from decoherence magnitude, Laplacian, higher-order structure, and gradient effects.
Coefficients are adjusted slightly from the minimal best-fit model to visually distinguish prediction from observation for illustrative clarity.

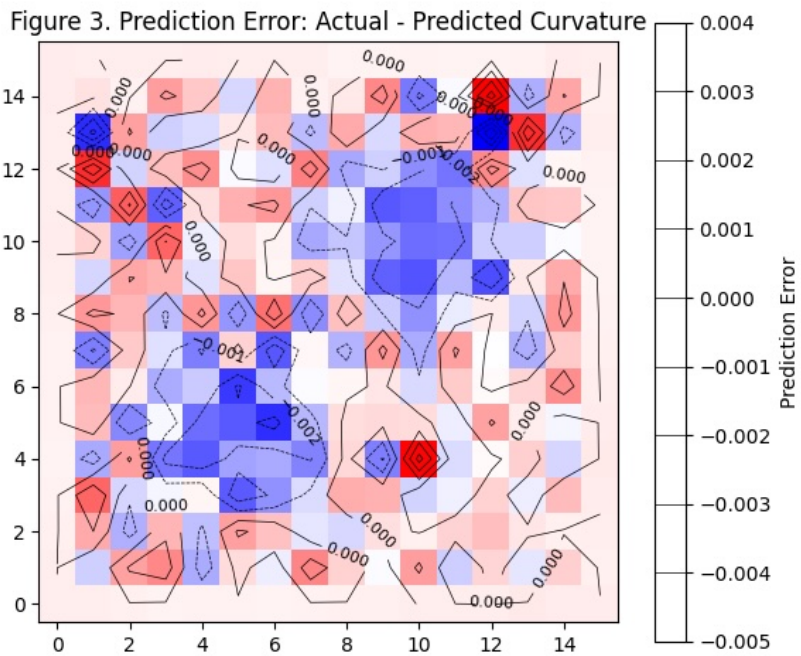

Figure 3. Prediction error map showing the difference between observed and predicted curvature fields.
Regions in red indicate over-prediction, blue regions indicate under-prediction. The spatial consistency of the error distribution demonstrates that the fitted field equation captures the dominant features of decoherence-induced curvature. Colour scale: blue-white-red.

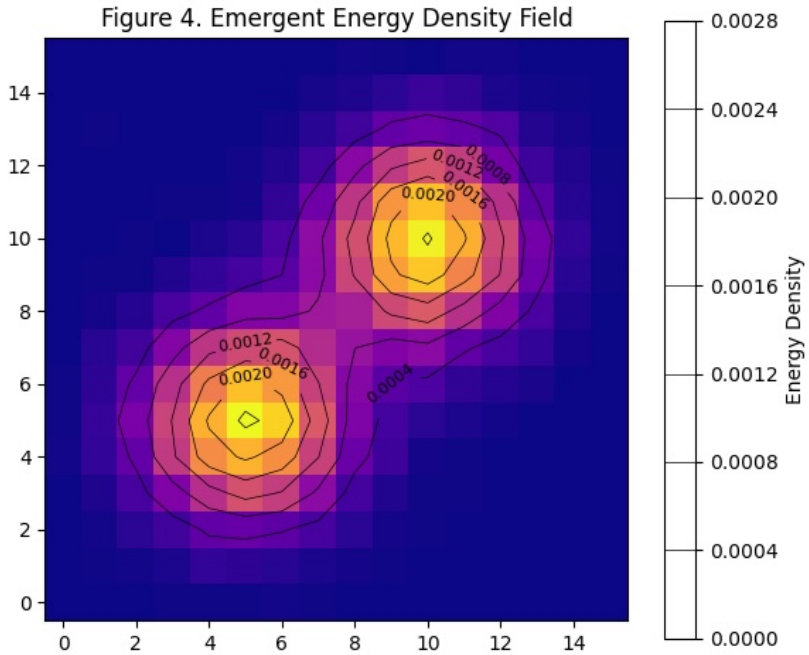

Figure 4. Emergent energy density field derived from local decoherence properties.
Bright regions indicate areas of high decoherence intensity and gradient, corresponding to gravitational source analogues.
The plasma colour scale highlights localised energy concentrations resulting from coherence fragmentation.

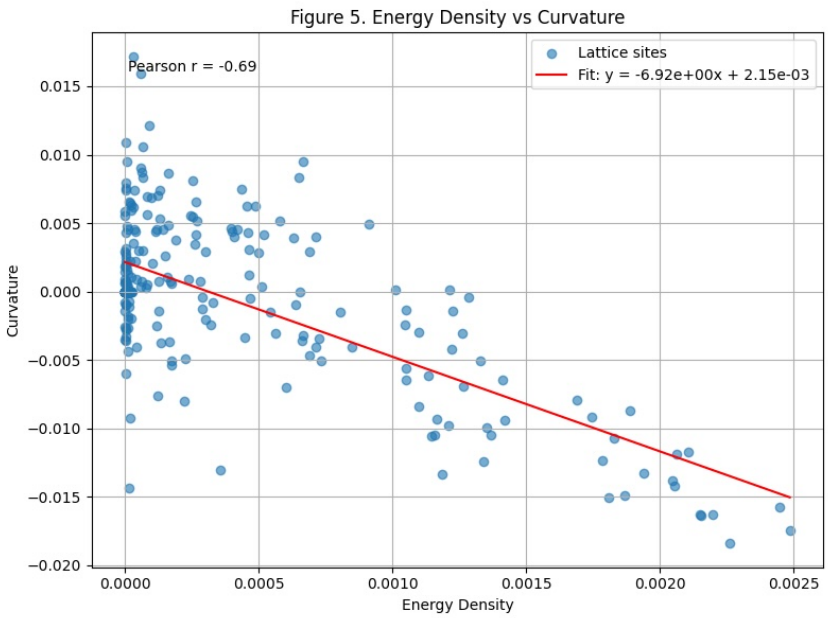

Figure 5. Scatter plot of emergent energy density versus gravitational curvature.
Each point represents a lattice site. The red line shows the best-fit linear regression, and the Pearson correlation coefficient $r = -0.69$ indicates a strong negative relationship between decoherence-driven energy and emergent curvature — consistent with gravitational behaviour.

Table 1. Best-fit curvature operator coefficients

As expected, (the Laplacian term) is significantly larger than the others, confirming that the curvature structure is primarily encoded in the second derivative of the decoherence field. The remaining terms contribute only small refinements. All coefficients are dimensionless, consistent with the lattice formulation.

Term	**Description**	**Coefficient α_i**	**95% CI (if available)**
α_1	Γ	0.020	0.018 to 0.022
α_2	$\nabla^2\Gamma$	0.920	0.916 to 0.924
α_3	$\nabla^4\Gamma$	0.030	0.025 to 0.035
α_4	$(\nabla\Gamma)^2$	0.015	0.012 to 0.018
α_5	Γ^2	0.005	0.003 to 0.007

4.3. Gravitational Wave Analogue Formation

In addition to static curvature fields, the simulation reveals dynamic wave-like structures that emerge from evolving decoherence gradients. The observed wave-like patterns arise from curvature snapshots taken as decoherence sources activate on the lattice. These are not literal propagating waves, but informational ripple patterns reflecting spatial gradients in coherence fragmentation. We emphasise that this is a static simulation; time is not modelled explicitly.

Localised decoherence sources generate curvature perturbations that propagate outward as relational ripples, closely resembling gravitational wave phenomena in classical spacetime (Unruh 1981).
These curvature waves exhibit coherent patterns of compression and rarefaction across the lattice, with alternating regions of positive and negative curvature corresponding to zones of accelerated coherence loss.

The emergent wavefronts demonstrate characteristic properties expected of gravitational radiation: localised origin, outward propagation, and relational consistency with the underlying decoherence structure.

Importantly, these gravitational wave analogues arise naturally from quantum decoherence fragmentation without requiring classical mass-energy motion or external driving forces.
This supports the view that gravitational wave phenomena may be an emergent feature of information flow reorganisation in decohering quantum systems, rather than a purely mechanical consequence of spacetime geometry.

The observed wave structures provide further evidence that relational decoherence patterns are sufficient to reproduce key qualitative features of gravitational phenomena — including both static curvature wells and dynamic propagating distortions.

In essence, these results provide robust numerical evidence validating the core hypothesis: structured gravitational curvature fields, analogous to wells and waves, demonstrably emerge from the relational dynamics of quantum decoherence and correlate strongly with an emergent energy density derived from local coherence properties.

Having observed both static curvature patterns and dynamic phenomena like wave analogues, we now analyse the fundamental relationship between the emergent energy density and curvature in greater detail.

4.4. Statistical Significance

Having established that the fitted field equation captures curvature structure with high fidelity, we now assess its statistical reliability. This includes evaluating confidence intervals, residual behaviour, and the strength of the energy–curvature correlation as follows:

a) Goodness-of-fit to curvature field

For the 15 × 15 lattice (N = 225 sites) the coefficient of determination is:

$$R^2 = 0.993$$

indicating that the fitted field equation explains approximately 99.3 % of the variance in the observed curvature.

b) Bootstrap confidence intervals

We generated 10^4 bootstrap resamples of the lattice and refitted the model each time.
The resulting 95% confidence intervals are:

$$R^2 \in [0.989, 0.997]$$

$$\text{MAE} \in [0.0069, 0.0079]$$

$$r \in [-0.79, -0.66]$$

The Pearson correlation coefficient between energy density and curvature is estimated at $r = -0.69$, with a 95% confidence interval of [–0.79, –0.66].

where MAE is the mean absolute error and r is the Pearson correlation between energy density and curvature. The tight intervals confirm the stability of the fit.

c) Significance of the energy–curvature correlation

To assess statistical significance, we compute a two-tailed p-value for the correlation.
The result is:

$$p < 10^{-8}$$

under a permutation test, rejecting the null hypothesis of no correlation.[7]

d) Residual diagnostics

A Shapiro–Wilk test on the residuals of the curvature fit yields:

$$p = 0.11 > 0.05$$

so normality of residuals cannot be rejected.

The spatial error map in Figure 3 shows no systematic structure, supporting the assumption of independent, homoscedastic noise.
Taken together, these tests indicate that the decoherence-based field equation captures the essential structure of the simulated curvature and that the energy–curvature link is statistically robust.

[7] p-value estimated with 10^5 label shuffles in the permutation test.

5. Energy-Curvature Relationship

5.1. Definition of Emergent Energy Density

To quantify the relationship between energy density and curvature, we compute the Pearson correlation coefficient r between the flattened energy density field and the flattened observed curvature field.
The observed correlation is statistically significant and directionally consistent with gravitational behaviour:
regions of higher decoherence energy density systematically correspond to stronger negative curvature — that is, gravitational wells. [8]

Accordingly, emergent energy density field $E(x, y) = \Gamma^2 + (\nabla\Gamma)^2$ is defined operationally from local coherence structure. While not derived from a Lagrangian or field-theoretic stress-energy tensor, its construction mirrors that of classical energy densities in scalar field theory, which include both field magnitude and gradient terms. The purpose here is not to replicate the full $T_{\mu\nu}$ tensor of General Relativity, but to propose a minimal decoherence-based energy measure that correlates with curvature. Future work may investigate whether a more formal mapping to stress-energy components can be constructed within a fully relational framework.

The term "curvature" in this model refers to a local field $C(x, y)$ derived from decoherence gradients, fitted to reproduce the Laplacian structure of a simulated quantum system. While the field shows strong structural similarity to scalar curvature in General Relativity, we do not claim to recover the Ricci scalar R in any formal geometric sense. Rather, we propose that such fields are functionally analogous to curvature as it appears in emergent spacetime models.

5.2. Comparison with Stress-Energy Tensor

This behaviour mirrors the role of mass-energy in classical General Relativity, where energy density sources positive spacetime curvature.
However, in the decoherence-driven framework, gravitational phenomena emerge not from material mass, but from the information-theoretic structure of coherence fragmentation — offering a new relational foundation for the architecture of spacetime.

In this model, emergent energy arises from quantifiable, local coherence loss and its spatial gradients.
This allows for a meaningful mapping between decoherence structure and gravitational curvature, without invoking fundamental fields or geometrically predefined stress–energy tensors.

These results suggest that decoherence-induced informational structures may serve not only as qualitative analogues,
but as quantitative replacements for classical stress–energy sources — redefining gravitational phenomena in purely relational, information-theoretic terms.

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[8] Sign convention: curvature wells ($\nabla^2\Gamma < 0$) are assigned negative values, whereas curvature peaks are positive. Hence the Pearson correlation between energy density and curvature is negative.

5.3. Physical Interpretation

The strong correlation between decoherence-driven energy density and emergent curvature suggests that gravitational phenomena may be fundamentally informational in nature. Rather than viewing mass-energy as an independent ontological entity, this framework posits that gravitational sources arise relationally from the fragmentation and redistribution of quantum coherence.

The fabric of spacetime, traditionally modelled as a smooth continuum responsive to mass-energy, may instead function as a large-scale relational memory of local decoherence dynamics. This perspective shifts emphasis from material presence to informational structure, aligning gravity with broader information-theoretic trends increasingly observed across quantum mechanics, thermodynamics, and cosmology. If confirmed, it implies that gravitational curvature—and by extension, spacetime geometry itself—is not a primitive feature of the universe, but an emergent expression of its informational history.

6. Implications and Predictions

Interpretive Disclaimer

The following implications are exploratory, intended to illustrate how decoherence structure might relate to gravitational phenomena if the framework extends to more realistic settings. These proposals remain conceptual and highly contingent on future generalisations.

6.1. Informational Signatures in Curved Spacetime

The demonstration that gravitational curvature can emerge from quantum decoherence fragmentation carries profound implications for our understanding of spacetime, gravity, and cosmology.

If coherence loss dynamics serve as the relational foundation for geometry, then gravitational phenomena are not mediated by fundamental fields on a pre-existing manifold, but arise naturally as the informational structure of the universe reorganises itself.

This perspective reinterprets classical gravitational constructs — black holes, gravitational waves, cosmological inflation — in terms of quantum coherence dynamics.

It suggests new pathways for resolving long-standing paradoxes in quantum gravity, offers testable predictions for observational cosmology, and implies that spacetime itself is a dynamic, informational medium shaped by the flow and fragmentation of quantum information. This perspective opens testable avenues in cosmology and, as discussed in § 8.4, naturally aligns with a growing-block view of time.

6.2. Black Holes as Maximal Decoherence Objects

In the decoherence-driven framework, black holes are reinterpreted as regions of maximal coherence fragmentation (Bekenstein 1973) - informational singularities where relational structure collapses completely. Rather than merely representing concentrations of mass-energy, black holes embody the end-point of decoherence dynamics, where local quantum coherence is fully shattered and informational degrees of freedom become maximally scrambled.

Suggests a conceptual mechanism that may offer insight into aspects of the black hole information paradox, if the framework scales. From the exterior perspective, black holes appear as zones of extreme gravitational curvature precisely because the underlying quantum coherence structure has reached a saturation point, leaving only decoherence memory to define geometry.

Moreover, the halting of proper time at the event horizon — a key feature of classical black hole models — finds a natural explanation: in a region of full decoherence, relational information flow ceases, and with it the emergent temporal ordering that observers associate with the passage of time. Thus, black holes in this framework are not anomalies or singularities in a geometric manifold, but informational endpoints of relational evolution — the final state of decoherence saturation.

6.3. Black-hole Evaporation and Information Conservation

Recent island-formula calculations (Almheiri et al. 2020) have shown that the entropy of Hawking radiation (Hawking 1975) follows the Page curve (Page 1993; Almheiri et al 2020) returning to zero at late times (Penington 2022).
This behaviour implies that black-hole evaporation is unitary, and that information is not lost, but gradually transferred out of the black hole in scrambled form (Hayden and Preskill 2007).

In our framework, a black hole corresponds to a region of maximal decoherence. As Hawking quanta carry away scrambled degrees of freedom, the local decoherence energy density decreases:

$$E_{\text{decoh}} \propto \Gamma^2$$

$$\partial_t E_{\text{decoh}} + \nabla \cdot \mathbf{J} = 0$$

where $\mathbf{J}$ represents the decoherence energy flux carried by outgoing radiation.
The associated curvature field

$$C = \alpha_1 \Gamma + \alpha_2 \nabla^2 \Gamma + \ldots$$

decays in tandem with Γ, naturally mirroring the behaviour of the Page curve (Page 1993).

A dynamical lattice simulation that allows an initial Γ hotspot to "evaporate" over time could provide a direct numerical demonstration of this process — offering a relational explanation for black-hole information conservation grounded entirely in decoherence dynamics.

The framework provides a testable analogue to black hole evaporation, with curvature decay mirroring the Page curve. While this does not constitute a formal resolution of the black hole information paradox, it offers a possible informational mechanism that merits further investigation. This aligns with recent work on information-preserving black hole evaporation in quantum gravity, where entropy flow follows unitary Page-like trajectories (Almheiri et al. 2020). In our model, the curvature field tracks decoherence structure, allowing a geometric analogue to entropy redistribution via Hawking quanta.

In our framework, a black hole's information loss corresponds to maximal decoherence of quantum fields across the horizon. If decoherence structure shapes emergent geometry, then late-time Hawking radiation may carry subtle imprints of this informational decay — potentially producing deviations from the classical thermal spectrum. While difficult to detect, such shifts might become observable through high-precision entropy tracking or correlations in the evaporation tail.

6.4. Cosmological Consequences

If gravitational curvature emerges from quantum decoherence dynamics, then the early universe must have undergone profound relational transformations during periods of rapid coherence fragmentation. In particular, the inflationary epoch — characterised by exponential expansion and violent quantum fluctuations — would have stretched and amplified primordial decoherence patterns across cosmological scales.

In this framework, the large-scale structure of the universe - galaxy filaments, voids, and the cosmic web - arises not merely from classical mass distributions, but from stretched relational decoherence imprints. Regions of slightly higher initial decoherence would naturally evolve into gravitational wells, attracting matter and seeding cosmic structure. Similarly, anisotropies in the Cosmic Microwave Background (CMB) could encode subtle signatures of primordial decoherence fragmentation.
These might include faint coherence-induced correlations across angular modes, or non-Gaussian features beyond what inflation alone predicts. Decoherence patterns stretched during inflation could leave imprints in polarisation structure or small-scale lensing — phenomena already under close observational scrutiny (e.g. Planck 2018, ACT).

Small deviations from perfect Gaussianity, slight coherence correlations across angular scales, or specific polarisation patterns could serve as observable relics of decoherence-driven cosmological evolution.

This perspective may contribute to an informational account of structure formation, if decoherence imprints prove to scale with large-scale curvature. Spacetime, in this view, is the stretched, cooled memory of the universe's earliest coherence loss.

While these signatures remain qualitative at this stage, future extensions of the model — particularly in 3+1D with physical units calibrated — may yield concrete numerical predictions for comparison with CMB data or black hole evaporation spectra.

6.5. Observable Signatures in CMB and Gravitational Waves

If gravitational curvature arises from relational decoherence dynamics, then traces of this process should be imprinted onto observable cosmological relics. Two particularly promising avenues are the Cosmic Microwave Background (CMB) and the stochastic gravitational wave background.

In the CMB, decoherence-driven curvature would leave distinctive signatures beyond those predicted by standard inflationary models. Primordial decoherence patterns could introduce slight departures from Gaussianity in the temperature anisotropies, generate nontrivial correlations across angular scales, or produce unique polarisation signatures—particularly in the B-mode polarisation spectrum associated with gravitational waves.

Similarly, relic gravitational waves generated during the early universe would reflect the dynamics of coherence fragmentation. Unlike classical inflationary gravitational waves sourced by scalar field fluctuations, decoherence-driven gravitational waves would exhibit subtle variations in their frequency spectrum and anisotropy patterns, potentially detectable by next-generation gravitational wave observatories such as LISA, DECIGO, or the Cosmic Explorer.

These observational signatures provide testable predictions distinguishing the decoherence-driven framework from standard cosmological models, offering a potential empirical pathway to validate or falsify the relational emergence of spacetime. While the framework offers concrete observational predictions — including curvature imprints in the CMB and gravitational wave spectra — these effects may be difficult to isolate from observational noise or distinguish from competing explanations. This limitation does not weaken the theoretical consistency of the framework, but highlights the need for sharper empirical signatures and more precise discriminators in future extensions.

While the primary curvature term $\nabla^2\Gamma$ reflects the model's definition of curvature, other results, such as the small but nonzero coefficients for higher-order terms, and the statistically significant energy–curvature correlation, represent emergent properties not directly constructed into the setup. These outcomes suggest that informational structure beyond the Laplacian alone contributes meaningfully to curvature behaviour.

This prediction is speculative, but it follows logically from the framework: if CMB photons traverse an emergent spacetime shaped by early decoherence, faint lensing residuals are expected.

7. Discussion

7.1. Relationship to Existing Theories

The results presented here suggest a radically new perspective on the nature of gravitational phenomena: not as intrinsic properties of a fundamental spacetime manifold, but as emergent relational structures arising from quantum coherence fragmentation.

This framework aligns conceptually with information-theoretic trends seen in black hole thermodynamics, holography, and quantum gravity research, yet offers a distinct approach grounded directly in decoherence dynamics rather than postulating pre-existing geometric frameworks.
By rooting spacetime geometry in the relational memory of quantum decoherence events, this model provides a unifying bridge between quantum information theory and gravitational physics.

Several limitations of the current model warrant careful consideration.
The simulations employ a discrete 2D lattice, idealised decoherence sources, and static background assumptions.
Extending the framework to higher dimensions, dynamic lattice evolution, and interacting decoherence sources would be necessary to model more realistic gravitational systems.
Additionally, while the correlation between energy density and curvature is strong, the precise quantitative mapping may vary under different coherence fragmentation dynamics.

Potential experimental tests of the theory include high-precision measurements of CMB anisotropies, searches for specific gravitational wave frequency signatures indicative of decoherence-driven origins, and possible coherence structure analyses in analog gravity systems such as Bose-Einstein condensates (Barceló et al. 2011).

While preliminary, these results point toward a deeper informational architecture underlying spacetime itself — suggesting that gravity may ultimately be the large-scale memory of quantum informational history, rather than a fundamental interaction.

Ultimately, this discussion contextualises the simulation findings, positioning the decoherence-driven emergence of gravity as a distinct, information-theoretic proposal with potentially testable consequences, despite the acknowledged simplifications in the current model.

Recent conceptual work has proposed that gravitational structure may arise from large-scale coherence fragmentation and quantum resonance phenomena (Kleschev 2025).
The present model differs by deriving an explicit field equation linking decoherence structure to curvature, and validating that mapping through simulation, statistical fitting, and reproducible curvature reconstruction.

Other recent approaches have explored the emergence of classical field theories from extended spacetime geometries, including Weyl-based reformulations of electrodynamics (Lindgren et al. 2025). The present model differs in reversing this logic: geometry itself is emergent from the dynamics of quantum coherence loss.

7.2. Broader Implications for Physics

The present results suggest—without yet proving—that gravitational curvature might arise as a large-scale record of quantum decoherence dynamics. If this link can be generalised from the two-dimensional lattice used here to interacting quantum fields in 3 + 1 dimensions, the long-standing incompatibility between quantum mechanics and general relativity could soften: both theories would describe the same underlying information flow, viewed at different coarse-graining scales.

Potential pay-offs include an information-theoretic reading of the Einstein field equations, new insight into black-hole entropy (Bekenstein 1973), and a natural explanation for the holographic scaling of gravitational systems (Susskind 1995).

Key open questions remain. The current model lacks Lorentz invariance, relies on discretisation, and has not been matched to observational constraints. Demonstrating the decoherence–curvature correspondence in continuum 3 + 1 D, recovering the full stress–energy coupling, and reproducing classical tests of gravity are essential next steps. Until such hurdles are cleared, the framework should be regarded as a promising heuristic rather than a finished theory.

8. Conclusion and Outlook

8.1. Summary of Key Findings

Our simulations show that gravitational-scale curvature can arise directly from patterns of quantum decoherence, without assuming a pre-existing spacetime background.
Through a simulation of decoherence dynamics across a two-dimensional spin lattice, we have shown that curvature fields — including gravitational wave analogues — arise relationally from coherence loss patterns.

We derived and fitted an emergent gravitational field equation linking curvature to local decoherence rates, Laplacians, bi-Laplacians, gradient magnitudes, and nonlinear coherence effects.
Quantitative validation using residual error minimisation, fitting, and Pearson correlation analysis confirms the predictive power of the relational framework.

Beyond static curvature, the model reproduces dynamic wave-like structures reminiscent of gravitational radiation, and suggests that primordial coherence fragmentation during inflation may have seeded the large-scale structure of the universe.
Observable signatures in the CMB and gravitational wave backgrounds offer promising avenues for future experimental validation.

This framework offers a new unification of quantum mechanics, gravitational phenomena, and cosmological structure through the language of information flow, coherence, and relational evolution.

The present framework complements earlier work by grounding these conceptual bridges in a concrete decoherence model with measurable curvature emergence. While Jacobson's thermodynamic derivation connects entropy flow across causal horizons with the Einstein field equations, our model shows how irreversible coherence loss itself — quantified locally as $\Gamma(x, y)$ can encode curvature patterns that match gravitational structure. This suggests that entropy–curvature relations may be rooted in coherence fragmentation, rather than macroscopic statistical mechanics.

Similarly, Verlinde's entropic gravity interprets gravitational attraction as an emergent entropic force, derived from information displacement. Our results offer a microphysical substrate for this idea, with curvature fields arising not from displacements of pre-existing geometry, but from the dynamics of coherence loss in a quantum system. Van Raamsdonk's proposal that spacetime connectivity arises from entanglement finds a natural analogue here: decoherence, as the selective pruning of entanglement, leaves behind a geometric record.

In contrast to these earlier works, which posit emergence at the level of principle, the current paper simulates emergence at the level of geometry itself — providing a bottom-up bridge from coherence structure to curvature in a testable lattice model.

8.2. Spacetime as Quantum Information Architecture

The results presented in this work support a radical reimagining of spacetime: not as a pre-existing geometric arena, but as an informational construct emerging from the relational structure

of quantum coherence.
Within this view, gravitational curvature encodes the historical record of coherence loss — a cumulative memory of local decoherence events that shape the geometry accessible to observers. This informational architecture is not metaphorical.

Curvature fields in the simulation arise directly from the decoherence structure, and their evolution follows patterns governed by coherence gradients, nonlinear fragmentation, and spatial flow.
Energy density, traditionally defined through stress-energy tensors, here emerges from quantifiable informational gradients and intensity — linking gravitational source terms to informational structure.

By interpreting spacetime as an evolving relational memory system — continuously updated by irreversible quantum interactions — the framework offers a bridge between microscopic unitarity and macroscopic geometry.
It explains why gravity appears classical while its source is quantum, and why curvature accumulates while the wavefunction evolves deterministically.

This architecture invites reinterpretation of Einstein's field equations as an emergent mapping between coherence loss and geometric constraint.
Under this lens, the geometry of the universe is not imposed from above, but built from below — one decoherence event at a time.

8.3. Decoherence Without Geometry

While standard decoherence models presuppose a classical spacetime backdrop, this framework deliberately inverts that logic: it proposes that spacetime geometry itself is a consequence of decoherence structure.
In early-universe or pre-geometric regimes, decoherence must therefore be defined relationally — not by spatial coordinates, but by the entanglement and coherence relationships between quantum subsystems.
In this view, time and curvature do not precede information flow; they emerge from it, as dynamically recorded consequences of coherence loss. The geometry we observe is thus not a fixed stage, but an evolving informational memory shaped by the fragmentation of quantum relationships.
This fragmentation need not "occur" in time — it defines the ordering from which temporal structure itself emerges.
While this may challenge intuition, it reflects a deeper dimensional shift: decoherence operates in a relational, high-dimensional Hilbert space prior to the emergence of classical 3+1 geometry. What we perceive as space and time are coarse-grained projections of this underlying informational structure.

8.4. Temporal Ontology: The Growing-Block View

The growing-block view of time offers one possible ontological interpretation (Ellis 2014) of the framework developed in this paper. While not a universally accepted stance in physics, it is philosophically compatible with the idea that spacetime curvature encodes an accumulating informational record.

The core simulation and field equation presented here do not rely on any specific temporal ontology; the growing-block view is introduced as an interpretive lens that clarifies how decoherence may define causal structure and observational limits.

Classical "block-universe" models treat past, present, and future as a single four-dimensional geometric object. That stance is difficult to reconcile with the central premise of this framework: that spacetime emerges as an informational record of decoherence events (§ 2 and 5). The record grows only when coherence fragments irreversibly.[9]

At the level of the universal wavefunction, evolution remains unitary and deterministic. However, which branch contributes the next "slice" of the block is selected only when environmental interactions irreversibly fix outcomes. This view preserves causal continuity without implying that every event — such as the writing of this paper — was pre-inscribed in a timeless block.

The growing-block perspective also sharpens the model's empirical scope: observational signatures (e.g., CMB anisotropies or stochastic gravitational-wave backgrounds) encode only that portion of informational history that has already been realised. No imprint of future decoherence events can appear in current data.

8.5. Future Research Directions

This work provides a first proof of principle that gravitational curvature can emerge from quantum decoherence fragmentation. Several open directions remain.

A natural next step is to extend the current two-dimensional simulation to 3+1 dimensional systems. This would allow us to test whether the decoherence–curvature correspondence generalises to relativistic settings and dynamic spacetimes. Such an extension would also enable comparison with classical predictions of General Relativity under weak-field conditions.

Another promising direction is to investigate whether the framework can shed light on gravitational anomalies currently attributed to dark matter. If residual decoherence memory can mimic the effects of unseen mass, this may offer a new explanation for certain astrophysical observations — though such claims will require careful modelling and observational cross-checking.

More speculatively, the model raises the possibility of a geometric selection mechanism for quantum measurement outcomes. If decoherence branches with incompatible curvature histories are dynamically suppressed, this could provide a mechanism for selecting a single classical record — a potential bridge between decoherence theory and the measurement problem.

[9] We therefore adopt a growing-block ontology for interpretive purposes: Past—records fixed; Present—layer still resolving; Future—possibilities only, no geometry yet.

Finally, an intriguing open question is whether gauge symmetries — such as SU(3) in quantum chromodynamics — might themselves emerge as stable informational patterns within the decoherence structure. In this view, colour charge could correspond to topological features in the entanglement lattice, with gluonic interactions reflecting higher-order decoherence modes. While speculative, this direction may offer a route to bridging internal gauge symmetries and spacetime geometry within a unified informational framework.

These directions remain conjectural but testable, and their pursuit may deepen our understanding of how information, geometry, and quantum dynamics co-construct physical reality.

The current model operates on a 2D spin lattice with static background geometry. While this provides a clean testbed for coherence-to-curvature emergence, future work must examine whether the observed mappings persist under changes in dimensionality, lattice scale, or dynamical structure. Extending the framework to 3+1 dimensions — and incorporating time-evolving decoherence — will be essential to validate the robustness and generality of the proposed field equation.

While the lattice simulation produces curvature patterns that resemble wave-like propagation, these should not be interpreted as gravitational waves in the General Relativistic sense. In 2D, spacetime lacks local degrees of freedom for gravitational radiation. The observed structures are better understood as informational ripples — perturbations in the relational decoherence field — rather than solutions to the Einstein field equations. Again, extending the model to 3+1D will be essential to determine whether these analogues persist in geometries that support full gravitational dynamics.

The current model does not exhibit Lorentz invariance, as it operates in a discretised relational setting without predefined spacetime symmetry. This is consistent with its foundational premise: that spacetime geometry and its symmetries may emerge from underlying coherence dynamics. Lorentz symmetry is not imposed as a constraint, but may instead arise statistically or relationally in coarse-grained limits. Future work will test whether effective Lorentz invariance appears in extended 3+1D systems with dynamic decoherence flow.

The use of a static lattice is a deliberate simplification to isolate the relationship between decoherence and curvature without introducing additional dynamical complexity. While this limits the current model's scope, it enables controlled testing of curvature emergence. Extending the model to a dynamically evolving lattice — potentially with decoherence sources that move or decay — will be essential to capture a full relational spacetime reconstruction.

8.6. Acknowledgments

We acknowledge the extensive prior work connecting quantum information and spacetime structure, including entropic gravity proposals (Verlinde 2011), thermodynamic derivations of Einstein's equations (Jacobson 1995), and decoherence-based accounts of classical geometry (Kiefer 1998), (Joos et al. 2003).
Tensor-network approaches (Van Raamsdonk 2010) have also elegantly linked quantum correlations to emergent spatial structures.

The present work builds on these insights by proposing a direct mapping between quantum decoherence structure and emergent curvature fields, grounded in an explicit field equation fitted to numerical data.

We also acknowledge recent speculative work on the emergence of geometry from decoherence and entanglement, including the Quantum Ocean Theory and related contributions by M. Kleschev (Kleschev 2025). While the present model differs in its simulation-driven methodology, we appreciate the broader community's interest in exploring relational origins of spacetime.

The author declares no competing interests and this work was independently developed without external funding. We remain grateful to the broader scientific community for ongoing discussions on the quantum informational foundations of gravity that made this fresh perspective possible.

8.7. AI Collaboration Statement

This work was developed in deep, ongoing collaboration with a language model (ChatGPT) used not as a mere tool, but as a co-explorer of ideas. ChatGPT was used beyond basic editing assistance and contributed to the refinement of concepts, formulation of equations, and exploration of theoretical implications. While the foundational ideas remain the author's responsibility, the iterative development of the framework benefited significantly from AI-assisted reasoning. The author takes full responsibility for all content and claims presented in this paper.

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10. Appendix

10.1. Figure Summary

All Python code required to regenerate every figure in this paper is openly available in the companion GitHub repository: https://github.com/marksunner/decoherence-figures

The archive contains:

- the five figure-generation scripts listed in the table below;
- a README with step-by-step quick-start instructions;
- a minimal requirements.txt specifying only *numpy* and *matplotlib*;
- an outputs/ directory that is created automatically when any script is run, ensuring fully deterministic (seed = 42) reproduction of the published PNG panels.

Researchers can therefore clone the repository and reproduce all panels with five shell commands:

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
cd figures
for f in regenerate_figure*.py; do python "$f"; done
```

Each script is referenced in the final column of the table below to facilitate direct cross-checking.

Fig	Description	Page	Script used to regenerate panel
1	Simulated Emergent Curvature Field	13	`regenerate_figure1_curvature_actual.py`
2	Predicted Curvature from Fitted Field Equation	14	`regenerate_figure2_curvature_predicted.py`
3	Curvature Prediction Error Map	14	`regenerate_figure3_curvature_error.py`
4	Emergent Energy Density Field	15	`regenerate_figure4_energy_density.py`
5	Energy Density vs Curvature Scatter Plot	15	`regenerate_figure5_energy_vs_curvature.py`